

AI and Machine Learning Intersection of AI and Physics

From Neural Networks to Physics-Informed AI

Department of Physics
Pusan National University

Spring Semester 2026

Level: 2nd Year Undergraduate

Contact Hours: 3 hours per week (2 hours lecture + 1 hour lab)

Tools: Python, Claude AI, VS Code

Team Project Information

- **Team Structure:** 4 students per team
- **Project Format:** Collaborative team project
- **Presentation:** End-of-semester team presentations
- **Collaboration Tools:** GitHub for code sharing and version control

1 Course Description

This course develops essential computational competencies in physics by integrating artificial intelligence and machine learning technologies in response to the demands of the AI era. The curriculum covers foundational Neural Networks, state-of-the-art Large Language Models (LLMs), and physics-specialized AI through Physics-Informed Neural Networks (PINNs).

Key Features

- Systematic theoretical learning based on MIT course materials
- "Vibe coding" practice utilizing Claude AI
- AI-based numerical solutions for mechanics, electromagnetism, quantum mechanics, and statistical physics
- PINN techniques directly incorporating physical laws into learning

2 Course Objectives

The course pursues the following goals:

1. Introduction to programming methodologies in the AI era
2. Understanding fundamental principles and operations of Neural Networks
3. Learning Deep Learning and Transformer architectures

4. Comprehensive understanding of AI (LLM-based) and machine learning principles
5. Developing efficient coding skills using LLMs
6. Application of these methods to solve problems in classical mechanics, electromagnetism, statistical physics, quantum mechanics, and related fields
7. Gaining practical experience in solving physics problems with AI
8. Mastering differential equation solutions using PINNs

3 Textbooks and References

3.1 Primary Textbook

- **MIT 6.S191:** Introduction to Deep Learning (lecture notes and videos)
- Instructor-provided materials: Jupyter Notebooks, Python code examples

3.2 Reference Books

- *Deep Learning* by Goodfellow, Bengio, and Courville
- *Physics-Informed Neural Networks* (collection of related papers)
- *Computational Physics* by Mark Newman

3.3 Online Resources

- Claude AI Documentation
- PyTorch/TensorFlow Tutorials
- ArXiv papers on PINN applications

4 Course Schedule (16 Weeks)

4.1 Part I: Neural Networks & Deep Learning (Weeks 1-7)

Week 1: Course Introduction and Environment Setup

- Course objectives and evaluation methods
- Python environment setup: uv, Git, VS Code/Cursor
- Claude AI account creation and basic usage
- **Lab:** "Hello, Neural Network!" - first neural network implementation, ML basics through polynomial fitting
- **Key Concepts:** Development Environment, AI in Physics

Week 2: Machine Learning Fundamentals

- Three types of machine learning: supervised, unsupervised, and reinforcement learning
- Data preprocessing and feature extraction
- Loss functions and optimization
- **Lab:** Linear Regression, clustering, Gradient Descent visualization
- **Key Concepts:** Supervised Learning, Loss Functions, Gradient Descent
- **Assignment 1:** Linear and nonlinear regression analysis using experimental data

Week 3: Neural Network Fundamentals

- Perceptron and Multi-Layer Perceptron
- Activation Functions: ReLU, Sigmoid, Tanh
- Mathematical structure of Forward Propagation
- Universal Approximation Theorem
- **Lab:** Pure Neural Network implementation with Numpy
- **Key Concepts:** Neurons, Activation, Forward Pass
- **Reference:** MIT 6.S191 Lecture 1

Week 4: Learning with Physics Data

- Regression models using TensorFlow/Keras
- Overfitting and Underfitting
- Relationship between model complexity and performance
- **Lab:** 1D function approximation, projectile motion prediction, pendulum period learning
- **Key Concepts:** Regression with Neural Networks, Model Complexity
- **Assignment 2:** Applying NN to various physics problems

Week 5: Core Deep Learning Techniques

- Regularization: L1/L2, Dropout, Batch Normalization
- Overfitting vs. Underfitting
- Data Augmentation and Transfer Learning concepts
- **Lab:** MNIST handwriting recognition (introducing CNN)
- **Key Concepts:** Regularization, Generalization, CNN Basics
- **Reference:** MIT 6.S191 Lecture 2

Week 6: Transformers and Attention Mechanism

- Limitations of RNN and emergence of Attention
- Principles of Self-Attention, Transformer architecture analysis

- Positional Encoding
- **Lab:** Attention mechanism, Multi-Head Attention, Transformer Block implementation
- **Key Concepts:** Attention, Self-Attention, Transformers, Sequence Modeling
- **Reference:** "Attention Is All You Need" (Vaswani et al., 2017)

Week 7: Introduction to Large Language Models (LLM)

- Comparing GPT, BERT, and Claude architectures
- Pre-training and Fine-tuning
- Token, Embedding, Context Window
- RLHF (Reinforcement Learning from Human Feedback)
- **Lab:** Tokenization methods, GPT vs BERT comparison, Claude API learning
- **Key Concepts:** LLM Architecture, Tokenization, Prompting, RLHF
- **Midterm Project Presentation:** Part I learning summary and mini-project

4.2 Part II: LLM Vibe Coding for Physics (Weeks 8-12)

Week 8: Midterm Exam / Introduction to LLM-based Coding

- What is "Vibe Coding"? - Code generation with natural language
- Prompt Engineering techniques
- Efficient debugging with Claude
- **Lab:** Generating physics simulation code with natural language
- **Key Concepts:** Prompt Engineering, AI-Assisted Programming

Week 9: Classical Mechanics Problem Solving

- Numerical solutions for Newton's equations (Euler, RK4)
- Planetary motion simulation
- Pendulum motion and Chaotic Systems
- Lagrangian and Hamiltonian mechanics
- **Lab:** Euler vs RK4 comparison, planetary motion, chaotic pendulum, Lagrangian/Hamiltonian
- **Key Concepts:** ODEs, Numerical Integration, Chaotic Dynamics
- **Assignment 3:** Mechanics problem solving and visualization using LLM

Week 10: Electromagnetism Simulation

- Numerical solutions for Maxwell equations (FDTD)
- Electric/magnetic field calculation and visualization
- Laplace equation and electrostatic problems
- Electromagnetic wave propagation simulation
- **Lab:** Electric/magnetic field basics, Maxwell equations 1D/2D, electromagnetic wave animation
- **Key Concepts:** Maxwell Equations, FDTD, Vector Field Visualization
- **Assignment 4:** Electric field analysis for complex charge distributions

Week 11: Quantum Mechanics Simulation

- Numerical solutions for Schrödinger equation
- Wave function visualization
- Tunneling effect simulation
- Finite Well, Harmonic Oscillator
- **Lab:** Schrödinger equation, wave functions, quantum tunneling, quantum wells
- **Key Concepts:** Quantum Mechanics, Wave Functions, Tunneling
- **Assignment 5:** Quantum state analysis in various potentials

Week 12: Statistical Physics and Monte Carlo Simulation

- Monte Carlo methodology
- Ising model and phase transitions
- Metropolis-Hastings algorithm
- Thermodynamic property calculation
- **Lab:** Random Walk, π estimation with Monte Carlo, 1D/2D Ising models
- **Key Concepts:** Statistical Physics, Monte Carlo, Phase Transitions
- **Midterm Project Check:** Part II lab results sharing

4.3 Part III: Physics-Informed Neural Networks (Weeks 13-14)**Week 13: PINN Fundamentals (ODE Edition)**

- Physics-Informed Neural Networks concepts
- Incorporating differential equations into Loss Function
- Automatic Differentiation
- PINN vs. traditional numerical methods (RK4)
- **Lab:** Simple ODE, harmonic oscillator, damped oscillator, Lorenz system
- **Key Concepts:** PINN, Physics Loss, Boundary Conditions
- **Reference Papers:** Raissi et al. (2019), Karniadakis et al. (2021)

Week 14: PINN Applications - Partial Differential Equations

- 1D/2D Heat Equation
- 1D/2D Wave Equation
- Burgers equation (nonlinear PDE)
- Handling complex boundary conditions
- **Lab:** PINN basic structure, Heat/Wave Equation, Burgers Equation
- **Key Concepts:** PDEs, PINN for Spatial-Temporal Problems
- **Assignment 6:** PDE solution project using PINN

4.4 Part IV: Final Project (Weeks 15-16)**Week 15: PINN Applications II - Advanced Topics & Final Project Start**

- Inverse Problems with PINN
- Multi-fidelity Learning
- Transfer Learning in PINN
- AI applications in modern physics research
- **Final Project Start:** Team topic selection and planning

Example Project Topics:

1. Solving Schrödinger equation using PINN
2. Electromagnetic problems with complex boundary conditions
3. Applying AI to phase transition simulations
4. Physics quantity prediction based on experimental data
5. Inverse Problem solving

Week 16: Final Project Presentation and Final Exam

- Team project result presentations (20 minutes)
- Code review and feedback

- Final exam (theory + practical problems)
- Course summary and future learning directions

5 Grading Policy

Category	Weight	Details
Assignments	30%	6 weekly assignments (5% each)
Midterm Project	25%	Part I summary presentation
Final Project	25%	Team project (code + report + presentation)
Attendance & Participation	20%	Class participation and discussion

5.1 Assignment Submission

- Code submission through GitHub Repository
- Include execution methods and result analysis in README.md

5.2 Final Project Requirements

- **4 students per team**
- Collaboration through GitHub
- 10-15 page report (LaTeX recommended)
- **Team presentation:** 20-minute presentation + 10-minute Q&A
- Reproducible code and data
- Specify each team member's contribution

6 Lab Environment

6.1 Required Software

```
# Python 3.9 or higher
conda create -n compphys python=3.10
conda activate compphys

# Required packages
pip install numpy scipy matplotlib
pip install pandas seaborn
pip install jupyter notebook
pip install torch torchvision # PyTorch
pip install anthropic # Claude API

# Optional packages
pip install plotly # Interactive visualization
pip install sympy # Symbolic computation
```

6.2 VS Code Extensions

- Python
- GitHub Copilot (optional)
- LaTeX Workshop (for report writing)

6.3 Claude AI Setup

- Create Claude.ai account

- API Key issuance (for lab)
- Prompt Engineering practice

7 Weekly Reading Materials

Week 1-2

Newman, "Computational Physics", Chapter 1-2; Python for Physics tutorial

Week 3-5

MIT 6.S191 Lecture Notes 1-3; Goodfellow et al., "Deep Learning", Chapter 6

Week 6-7

"Attention Is All You Need" - Vaswani et al. (2017); GPT-3 Paper (Brown et al., 2020)

Week 9-12

Newman, "Computational Physics", Chapter 8 (ODEs); Landau & Lifshitz, Classical Mechanics (reference)

Week 13-14

Raissi et al., "Physics-informed neural networks" (2019); Karniadakis et al., "Physics-informed machine learning" (2021); Cuomo et al., "Scientific Machine Learning" (2022)

8 Prerequisites

Required

- General Physics I, II
- Classical Mechanics
- Electromagnetism I
- Basic Python programming

Recommended

- Quantum Mechanics I
- Mathematical Physics
- Statistical Physics

9 Useful Links

9.1 Course Materials

- [MIT 6.S191 Course Website](#)
- [Physics-Informed Neural Networks GitHub](#)

9.2 Online Resources

- Physics-Informed DeepONet
- [DeepXDE Library](#)

9.3 Paper Repositories

- [ArXiv: Physics + AI](#)
- [ArXiv: Machine Learning](#)

10 Assignment Details

Assignment 1: Data Fitting (Week 2)

Fit experimental data with various regression models and select the optimal model

Assignment 2: Optimizer Comparison (Week 4)

Compare performance of SGD, Momentum, Adam optimizers on the same problem

Assignment 3: Mechanics Simulation (Week 9)

Write complex mechanics problem simulation code using LLM

Assignment 4: Electromagnetic Visualization (Week 10)

Calculate electric field for given charge distribution and create visualization

Assignment 5: Quantum Simulation (Week 11)

Calculate and analyze quantum states in various potentials

Assignment 6: PINN Project (Week 14)

Solve selected PDE using PINN and compare with traditional methods

11 Learning Outcomes

Upon successful completion of this course, students will be able to:

1. Understand the mathematical principles of Neural Network operations
2. Possess efficient coding skills using LLMs
3. Gain practical experience in solving physics problems with AI
4. Master differential equation solutions using PINNs
5. Apply the latest AI technologies to physics
6. Collaborate and manage code through GitHub
7. Write research papers and deliver presentations

12 Course Philosophy

*A physicist is one who understands nature,
a computational physicist simulates nature,
and a physicist in the AI era learns and predicts nature.*

This course aims not merely to teach how to use AI tools, but to learn **new problem-solving approaches by integrating physical intuition with AI technology.**

13 Contact and Office Hours

Classroom:	TBA
Lab Room:	TBA
Office Hours:	Wednesday 14:00-16:00 (by appointment recommended)
Email:	TBA
Course GitHub:	TBA
Q&A:	Slack channel

14 Important Notes

1. **Academic Integrity:** Sources must be clearly cited when using AI tools
2. **Code Sharing:** Assignments are individual work; code copying is prohibited
3. **Attendance:** Attendance is very important as this is a lab-oriented course
4. **Required Materials:** Laptop required (for lab work)

15 Expected End-of-Semester Deliverables

15.1 Individual

- Portfolio of 6 assignments
- Personal GitHub Repository
- Collection of physics simulation codes

15.2 Team

- Final project report
- Presentation materials
- Open-source code (GitHub)

16 Additional Learning Resources (Optional)

16.1 Online Courses

- Andrew Ng's Machine Learning Course (Coursera)
- Fast.ai Practical Deep Learning
- Stanford CS231n (Convolutional Neural Networks)

16.2 YouTube Channels

- 3Blue1Brown (Neural Networks Series)
- Two Minute Papers
- Arxiv Insights

16.3 Books

- *Neural Networks and Deep Learning* by Michael Nielsen (free)
- *Dive into Deep Learning* by Zhang et al. (free)
- *Computational Physics* by Giordano & Nakanishi

17 Course Plan Changes

- This syllabus may be adjusted according to learning progress
- Important changes will be announced at least one week in advance
- Lab content may be adjusted based on student feedback

Closing Remarks

Through this course, you will master **essential AI tools for 21st-century physicists**. Beyond simply executing code, we expect you to **combine physical intuition with AI technology to achieve creative problem-solving**.

Let's make this an exciting semester together!

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